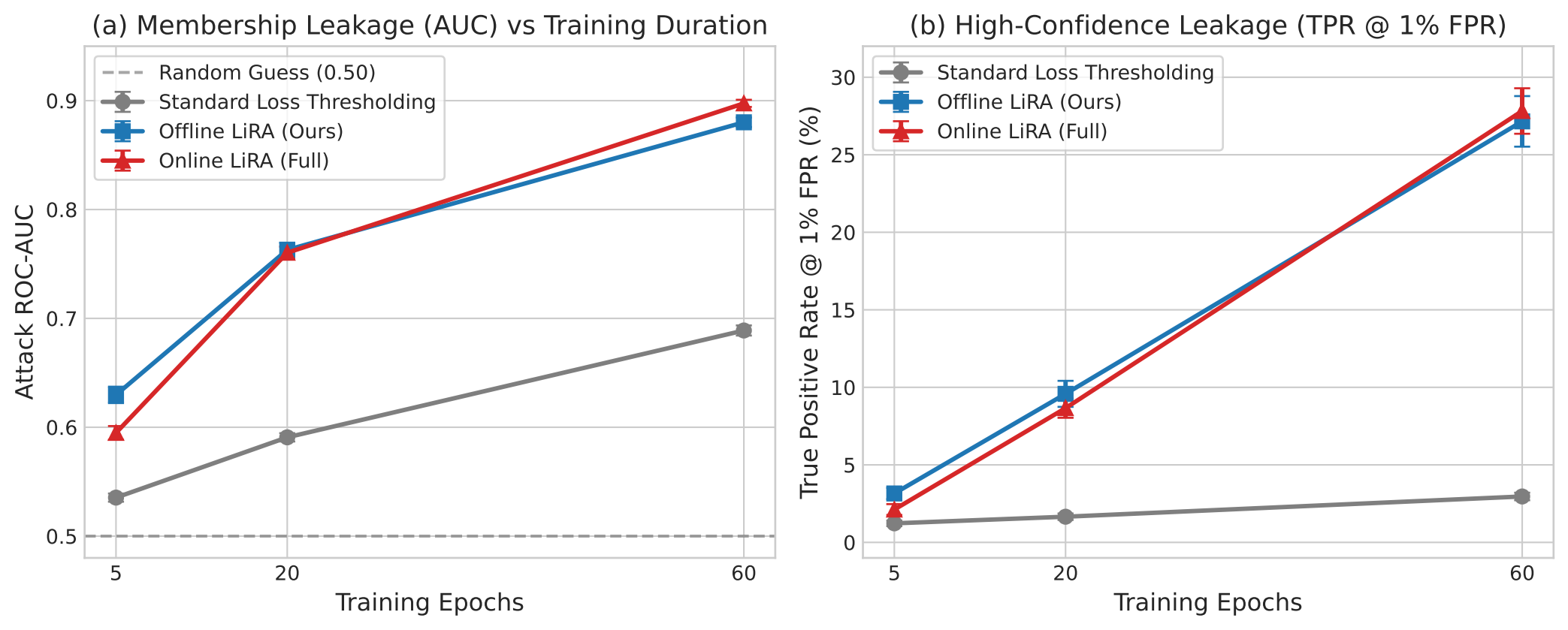
(a) Membership Leakage (AUC) vs Training Duration
Random Guess (0.50)
Standard Loss Thresholding
Offline LiRA (Ours)
Online LiRA (Full)
Attack ROC-AUC
0.9
0.8
0.7
0.6
0.5
5
20
60
Training Epochs
(b) High-Confidence Leakage (TPR @ 1% FPR)
Standard Loss Thresholding
Offline LiRA (Ours)
Online LiRA (Full)
True Positive Rate @ 1% FPR (%)
30
25
20
15
10
5
0
5
20
60
Training Epochs